

MATH1231 CALCULUS

Chapter 2A

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Topic 2 - Integration Techniques

powers of sine and cosine

multiple angles of sine and cosine

(powers of tan and sec or cosh and sin)



reduction formulae

substitutions

rational functions

Review

- Integration is the undoing of differentiation.

$$F(x) = \frac{df}{dx}$$

$$\int F(x) dx = \int \frac{d}{dx} f(x) dx = f(x) + C$$

- Chain Rule

$$\frac{d}{dx} f(g(x)) = g'(x) f'(g(x))$$

$$\int g'(x) f'(g(x)) dx = \int \frac{d}{dx} f(g(x)) dx = f(g(x)) + C$$

Review

What you should know already:

- ▶ using a table of integrals
- ▶ integration by inspection
- ▶ integration by parts
- ▶ integration by simple substitution

Integration Table (Examples)

TABLE Basic integration formulas

1. $\int du = u + C$	13. $\int \cot u \, du = \ln \sin u + C$ $= -\ln \csc u + C$
2. $\int k \, du = ku + C$ (any number k)	14. $\int e^u \, du = e^u + C$
3. $\int (du + dv) = \int du + \int dv$	15. $\int a^u \, du = \frac{a^u}{\ln a} + C$ ($a > 0, a \neq 1$)
4. $\int u^n \, du = \frac{u^{n+1}}{n+1} + C$ ($n \neq -1$)	16. $\int \sinh u \, du = \cosh u + C$
5. $\int \frac{du}{u} = \ln u + C$	17. $\int \cosh u \, du = \sinh u + C$
6. $\int \sin u \, du = -\cos u + C$	18. $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + C$
7. $\int \cos u \, du = \sin u + C$	19. $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$
8. $\int \sec^2 u \, du = \tan u + C$	20. $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1} \left \frac{u}{a} \right + C$
9. $\int \csc^2 u \, du = -\cot u + C$	21. $\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1} \left(\frac{u}{a} \right) + C$ ($a > 0$)
10. $\int \sec u \tan u \, du = \sec u + C$	22. $\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1} \left(\frac{u}{a} \right) + C$ ($u > a > 0$)
11. $\int \csc u \cot u \, du = -\csc u + C$	
12. $\int \tan u \, du = -\ln \cos u + C$ $= \ln \sec u + C$	

Integration Techniques

Procedures for Matching Integrals to Basic Formulas

PROCEDURE

Making a simplifying substitution

Completing the square

Using a trigonometric identity

Eliminating a square root

Reducing an improper fraction

Separating a fraction

Multiplying by a form of 1

EXAMPLE

$$\frac{2x - 9}{\sqrt{x^2 - 9x + 1}} dx = \frac{du}{\sqrt{u}}$$

$$\sqrt{8x - x^2} = \sqrt{16 - (x - 4)^2}$$

$$\begin{aligned} (\sec x + \tan x)^2 &= \sec^2 x + 2 \sec x \tan x + \tan^2 x \\ &= \sec^2 x + 2 \sec x \tan x \\ &\quad + (\sec^2 x - 1) \\ &= 2 \sec^2 x + 2 \sec x \tan x - 1 \end{aligned}$$

$$\sqrt{1 + \cos 4x} = \sqrt{2 \cos^2 2x} = \sqrt{2} |\cos 2x|$$

$$\frac{3x^2 - 7x}{3x + 2} = x - 3 + \frac{6}{3x + 2}$$

$$\frac{3x + 2}{\sqrt{1 - x^2}} = \frac{3x}{\sqrt{1 - x^2}} + \frac{2}{\sqrt{1 - x^2}}$$

$$\begin{aligned} \sec x &= \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} \\ &= \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \end{aligned}$$

Examples

$$\int 6x(1 + x^2)^2 dx =$$

$$\int \cosh x \cosh(\sinh x) dx =$$

$$\int 2e^{2x} \cos(e^{2x}) dx =$$

$$\int \tan(x) dx =$$

Substitution

$$I = \int f(x) dx$$

Put $x = g(u)$ then $dx = g'(u) du$ and $f(x) = f(g(u))$.

$$I = \int f(g(u))g'(u) du = \int h(u) du$$

This is only useful if $\int h(u) du$ is easier than $\int f(x) dx$.

Example

$$\int \frac{\cos x}{1 + \sin x} dx$$

Integration by parts

$$\frac{d}{dx}(u v) = \frac{du}{dx} v + u \frac{dv}{dx}$$

$$\int u \frac{dv}{dx} dx = u v - \int v \frac{du}{dx} dx$$

Example $\int x e^x dx$

Example

$$\int x^3 e^x dx$$

(2.1) Trigonometric Integrals

How do we obtain results such as the examples below?

$$\int \sin^3 x \cos^6 x \, dx = -\frac{1}{9} \sin^2 x \cos^7 x - \frac{2}{63} \cos^7 x + C$$

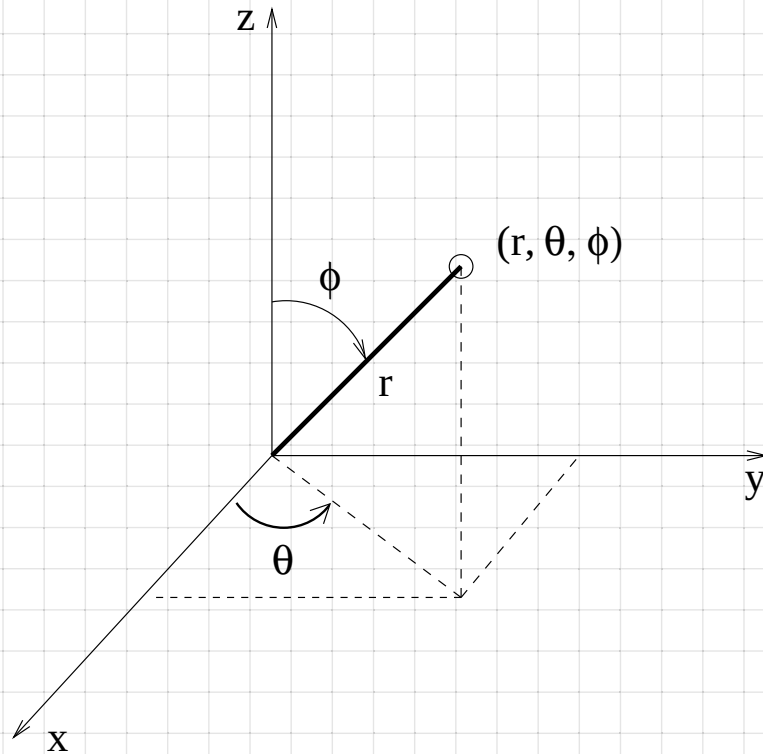
$$\begin{aligned} \int \cos^8 x \, dx &= \frac{1}{8} \cos^7 x \sin x + \frac{7}{48} \cos^5 x \sin x \\ &\quad + \frac{35}{192} \cos^3 x \sin x + \frac{35}{128} \cos x \sin x \\ &\quad + \frac{35}{128} x \end{aligned}$$

$$\begin{aligned} \int \tan^2 x \sec^3 x \, dx &= \frac{1}{4} \tan^3 x \sec x + \frac{1}{8} \tan^2 x \sin x + \frac{1}{8} \sin x \\ &\quad - \frac{1}{8} \ln(\sec x + \tan x) \end{aligned}$$

$$\int \sinh^3 x \cosh^2 x \, dx = \frac{1}{15} \cosh^3 x (3 \cosh^2 x - 5)$$

Who Cares?

Many problems in three dimensional space can be simplified using spherical polar co-ordinates



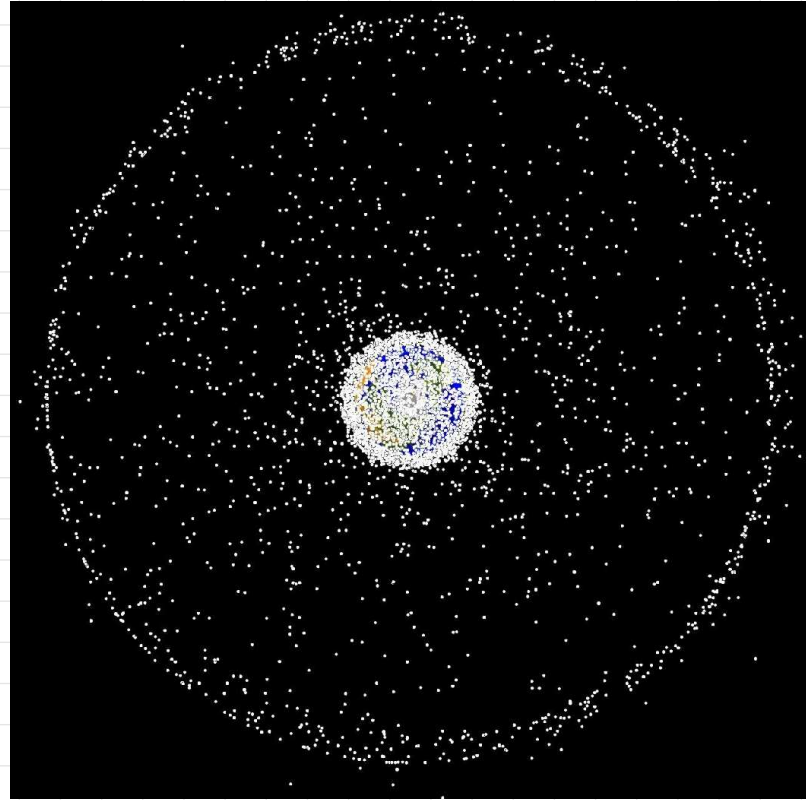
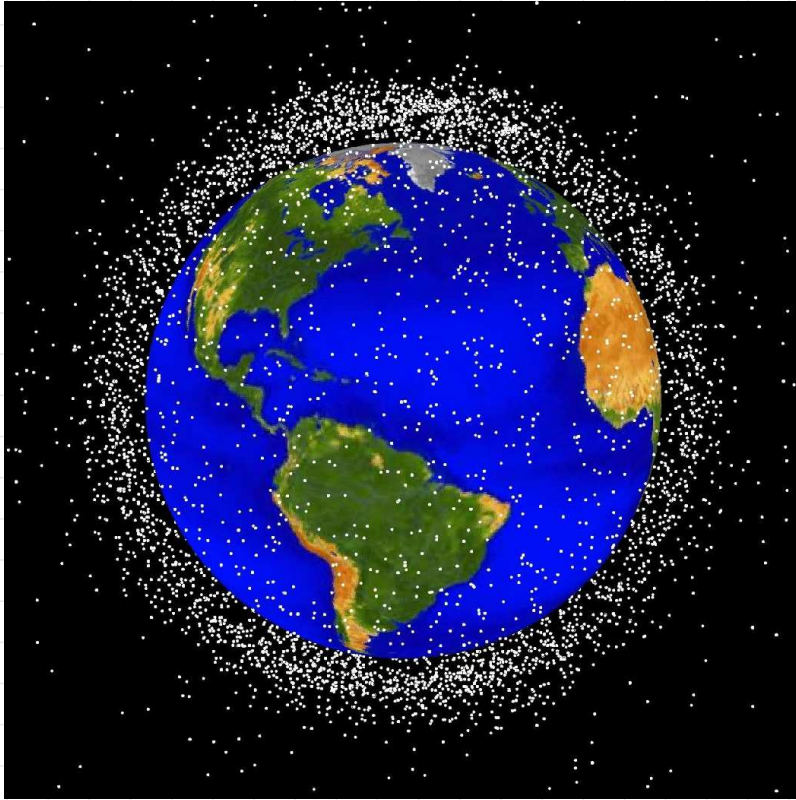
$$x = r \sin \phi \cos \theta$$

$$y = r \sin \phi \sin \theta$$

$$z = r \cos \phi$$

Integrals over powers of trig functions then arise, e.g., volume calculations, charge distributions, acoustic vibrations, angular momentum of electrons, trajectories of falling space junk

<http://orbitaldebris.jsc.nasa.gov>



Example

$$\int \cos^3 x \, dx$$

Example

$$\int \cos^3 x \sin^4 x \, dx$$

(2.1.1)

$$\int \cos^m x \sin^n x dx$$

look for an **odd** power

- If m is **odd** factor out $\cos x$

put $u = \sin x, \quad du = \cos x dx$

the remaining factor $\cos^{m-1} x$ is an even power $(m-1) = 2k$ use

$\sin^2 x + \cos^2 x = 1$ to eliminate **even** powers

$$\cos^{2k} x = (1 - \sin^2 x)^k = (1 - u^2)^k$$

integrate the polynomial in u

$$\int (1 - u^2)^k u^n du$$

- If n is **odd**, factor out $\sin x$ etc.
- If both m and n are odd, factor out either $\cos$ or $\sin$

Example

$$\int \sin^2 x \cos^2 x \, dx$$

$$\int \cos^m x \sin^n x \, dx$$

- If **both** $m = 2j$ and $n = 2k$ are **even**, we use the identities

$$(\cos^2 x)^j = \left(\frac{1 + \cos 2x}{2}\right)^j \quad \text{and} \quad (\sin^2 x)^k = \left(\frac{1 - \cos 2x}{2}\right)^k$$

to change the integral into a **sum** of integrals of the form

$$\int \cos^\ell 2x \, dx \quad \ell \in \mathbb{N}$$

if ℓ is **odd**, we use the method of the first section

if ℓ is **even**, we use the method of this section again.

(2.1.2)

$$\int \sin nx \cos mx \, dx$$

$$\int \sin nx \sin mx \, dx$$

$$\int \cos nx \cos mx \, dx$$

Use the identities

$$\sin A \cos B = \frac{1}{2} (\sin(A + B) + \sin(A - B))$$

$$\sin A \sin B = \frac{1}{2} (\cos(A - B) - \cos(A + B))$$

$$\cos A \cos B = \frac{1}{2} (\cos(A - B) + \cos(A + B))$$

Example

$$\int \sin 3x \sin 2x \, dx$$

(2.1.3)

$$\int \tan^m x \sec^n x \, dx$$

There are rules (see below), but don't worry about the detail of this for powers of $\tan$ and $\sec$. Look for something obvious or an easy substitution. Remember:

$$\sec^2 x = 1 + \tan^2 x, \quad \frac{d}{dx} \tan x = \sec^2 x, \quad \frac{d}{dx} \sec x = \tan x \sec x$$

Example

$$\int \sec^2 x \tan x \, dx$$

$$\int \tan^m x \sec^n x dx$$

- If n is **even**, factor out $\sec^2 x$

put $u = \tan x, \quad du = \sec^2 x dx$

use $\sec^2 x = 1 + \tan^2 x$ to eliminate remaining **even** powers of $\sec x$

integrate the polynomial in u

- If m is **odd**, factor out $\tan x \sec x$

put $u = \sec x, \quad du = \sec x \tan x dx.$

use $\sec^2 x = 1 + \tan^2 x$ to eliminate remaining **even** powers $\tan x$ etc.

Examples

$$\begin{aligned}\int \tan^2 x \overbrace{\sec^4 x}^{\text{even}} dx &= \int (\tan^2 x \sec^2 x) \overbrace{\sec^2 x}^{\text{factor}} dx \\&= \int \underbrace{\tan^2 x (1 + \tan^2 x)}_{\substack{\text{Put} \\ u = \tan x}} \underbrace{\sec^2 x}_{du} dx \\&= \int u^2 (1 + u^2) du = \frac{1}{3} u^3 + \frac{1}{5} u^5 + C \\&= \frac{1}{3} \tan^3 + \frac{1}{5} \tan^5 + C\end{aligned}$$

$$\begin{aligned}\int \overbrace{\tan^3 x}^{\text{odd}} \sec^3 x dx &= \int \tan^2 x \sec^2 x \overbrace{(\sec x \tan x)}^{\text{factor}} dx \\&= \int \underbrace{(\sec^2 x - 1) \sec^2 x}_{\substack{\text{Put} \\ u = \sec x}} \underbrace{\sec x \tan x}_{du} dx \\&= \int (u^2 - 1) u^2 du = \frac{1}{5} u^5 - \frac{1}{3} u^3 + C \\&= \frac{1}{5} \sec^5 - \frac{1}{3} \sec^3 + C\end{aligned}$$

$$\int \tan^m x \sec^n x dx$$

- If m is **even** $m = 2k$ and n is **odd**

$$\int \tan^m x \sec^n x dx = \int (\sec^2 x - 1)^k \sec^n x dx$$

yields a sum of integrals of the form

$$I(j) = \int \sec^j x dx$$

If j is even then factor our $\sec^2 x$ etc but what if j is odd?

- Use **Reduction Formulae!**

(2.2) Reduction formulae

- ▶ Repeated integration by parts can be a long process!
We can shorten the process by applying reduction formulae.
- ▶ Example:

$$\int \tan^n(x) dx = \frac{1}{n-1} \tan^{n-1}(x) - \int \tan^{n-2} x dx, \quad n \geq 2$$

Use this to evaluate $\int \tan^5(x) dx$:

Example

Given $I_n = \int \tan^n(x) dx$ prove the reduction formula

$$I_n = \frac{\tan^{n-1}(x)}{n-1} - I_{n-2}, \quad n \geq 2.$$

Example

Given $I_n = \int x^n e^x dx$ use integration by parts to derive a reduction formula

Example

Given $I_n = \int \sin^n x \, dx$ use integration by parts to derive a reduction formula

(2.2.3) A two-parameter reduction formulae (proof – see Notes)

$$I_{m,n} = \int_0^{\pi/2} \cos^m \theta \sin^n \theta d\theta = \begin{cases} \left(\frac{m-1}{m+n} \right) I_{m-2,n} & m \geq 2 \\ \left(\frac{n-1}{m+n} \right) I_{m,n-2} & n \geq 2 \end{cases}$$

with $I_{1,1} = \frac{1}{2}$, $I_{1,0} = 1$, $I_{0,1} = 1$, $I_{0,0} = \frac{\pi}{2}$

Example: evaluate $\int_0^{\frac{\pi}{2}} \cos^3 x \sin^4 x dx = I_{3,4}$

Sample Exam Question

Given $I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta \sec \theta d\theta$, show that $I_n = \frac{1}{n} (\sqrt{2} - (n-1)I_{n-2})$ for $n \geq 2$. (Hint $\sec \theta \tan \theta = \frac{d}{d\theta} (\sec \theta)$)

(2.3) Trigonometric and hyperbolic substitutions

Substitution is inspired guesswork. This table is for inspiration.

integral factor	substitutions		identities
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$ $dx = a \cos \theta d\theta$	$x = a \tanh \theta$ $dx = a \operatorname{sech}^2 \theta d\theta$	$\cos^2 \theta + \sin^2 \theta = 1$ $\operatorname{sech}^2 \theta + \tanh^2 \theta = 1$
$\sqrt{a^2 + x^2}$	$x = a \sinh \theta$ $dx = a \cosh \theta d\theta$	$x = a \tan \theta$ $dx = a \sec^2 \theta d\theta$	$\sec^2 \theta = \tan^2 \theta + 1$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$ $dx = a \sec \theta \tan \theta d\theta$	$x = a \cosh \theta$ $dx = a \sinh \theta d\theta$	$\cosh^2 \theta = \sinh^2 \theta + 1$

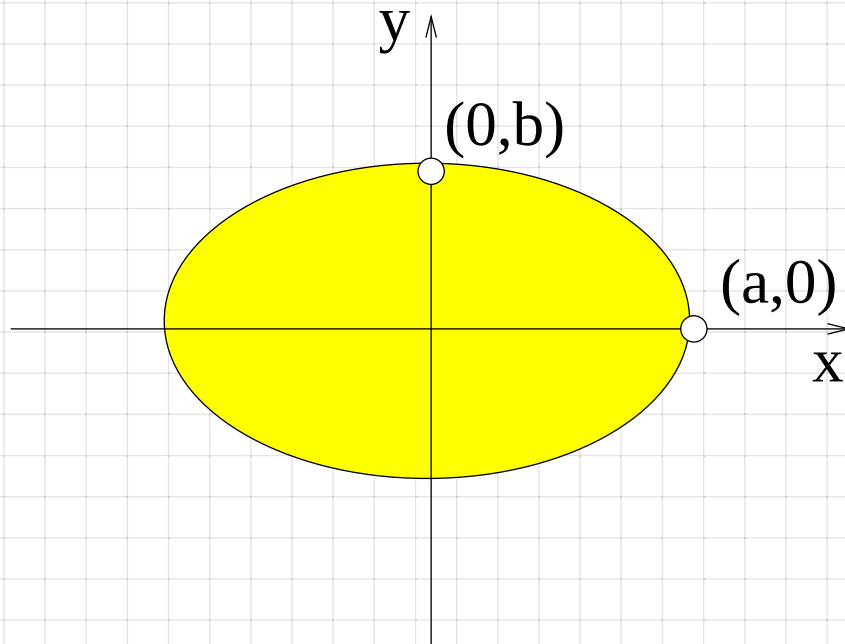
Example

Evaluate $I = \int \sqrt{9 - x^2} \, dx$

Example

Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$A = 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$$



Example

Evaluate $I = \int \frac{x^3}{\sqrt{x^2-16}} dx$

Example

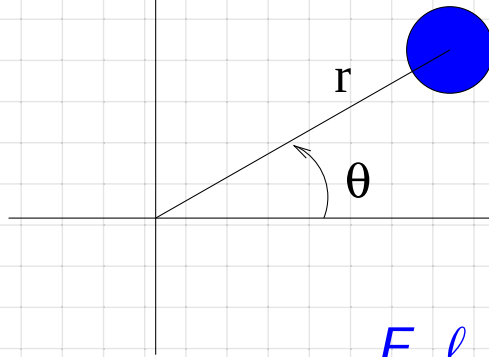
Evaluate $I = \int \frac{x dx}{\sqrt{4+x^2}}$ using

- (i) a non-trig. substitution
- (ii) a trigonometric substitution
- (iii) a hyperbolic substitution

Planetary Motion Ellipses – Kepler (1609,1619)

Newton (1687) – planets move under the influence of a gravitational force proportional to the square of the distance between the planet and the sun

Johann Bernoulli (1710)



$$\theta(r) = \int \frac{\frac{\ell}{r^2} dr}{\sqrt{2\mu(E + \frac{k}{r} - \frac{\ell^2}{2\mu r^2})}} + C$$

E, ℓ, μ, k, C are constants

This looks like a tricky integral, but with the right substitutions it simplifies nicely: First substitute $u = \frac{1}{r}$, $du = -\frac{1}{r^2} dr$.

$$\theta(r) = \int \frac{-\ell du}{\sqrt{2\mu(E + ku - \frac{\ell^2}{2\mu} u^2)}}$$

Example ctd.

Complete the square in the denominator

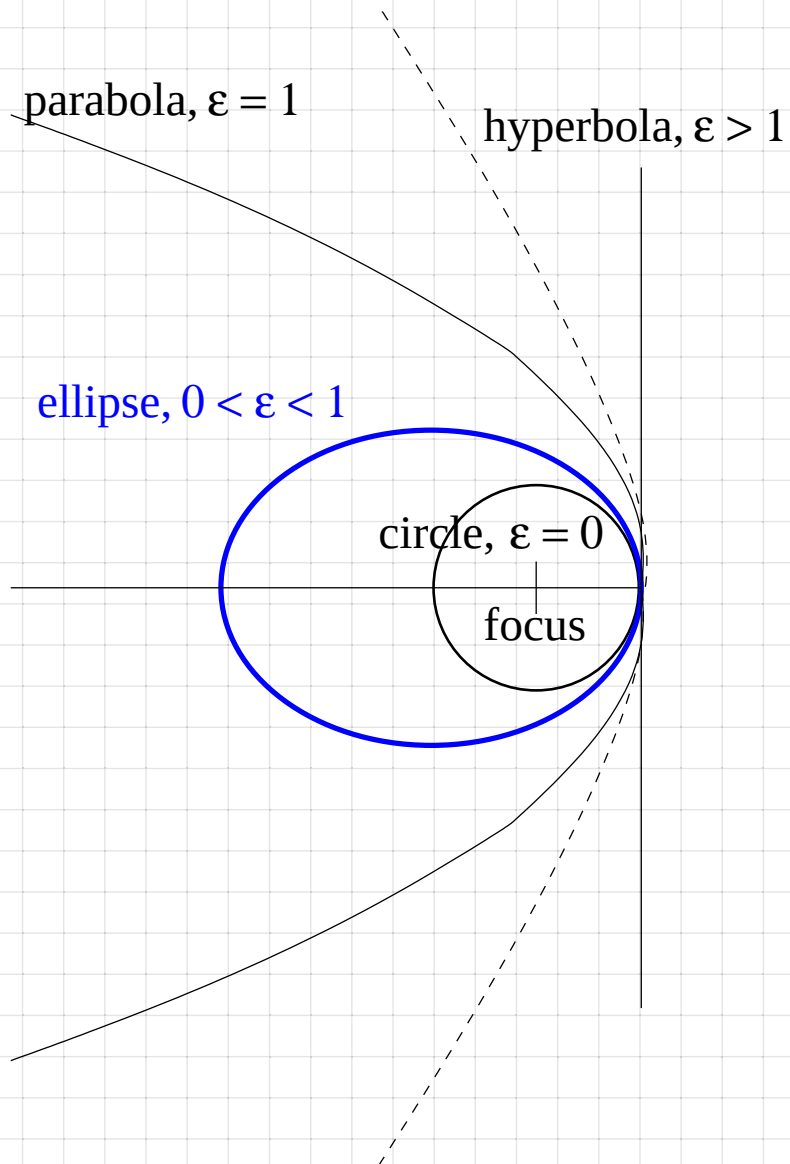
$$\theta(r) = \int \frac{-du}{\sqrt{\left(\frac{k^2\mu^2}{\ell^4} + \frac{2\mu E}{\ell^2}\right) - \left(u - \frac{k\mu}{\ell^2}\right)^2}}$$

Substitution

$$\left(u - \frac{k\mu}{\ell^2}\right) = \sqrt{\left(\frac{k^2\mu^2}{\ell^4} + \frac{2\mu E}{\ell^2}\right)} \cos \phi, \quad du = -\sqrt{\left(\frac{k^2\mu^2}{\ell^4} + \frac{2\mu E}{\ell^2}\right)} \sin \phi \, d\phi$$

$$\theta(r) = \int \frac{\sqrt{\left(\frac{k^2\mu^2}{\ell^4} + \frac{2\mu E}{\ell^2}\right)} \sin \phi \, d\phi}{\sqrt{\left(\frac{k^2\mu^2}{\ell^4} + \frac{2\mu E}{\ell^2}\right)} \sin \phi} = \phi$$

$$\theta(r) = \cos^{-1} \frac{\left(\frac{\alpha}{r} - 1\right)}{\epsilon}$$



$$\frac{\alpha}{r} = 1 + \epsilon \cos \theta$$

$$\alpha = \frac{\ell^2}{\mu k}$$

$$\epsilon = \sqrt{1 + \frac{2E\ell^2}{\mu k^2}}$$

Earth $\epsilon = 0.0167$

Halley's Comet $\epsilon = 0.967$